

A Modern E-commerce Platform Built with Django



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Python Web Development

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01 Project Overview

A fully-featured, well-architected modern e-commerce website built on the latest Django 5.2 framework. It implements core e-commerce functionalities such as product management, user authentication, shopping cart, order processing, and payment simulation.



Product Management

Supports unlimited category levels, multi-specification (color/size) products, inventory management, and automatic SKU generation.



Shopping Cart System

Session-based cart supporting add, modify quantity, and delete items.



Payment Simulation

Integrates simulated credit card payment, recording transaction status and desensitized information.



User Center

User registration, login, profile management, and favorites functionality.



Order System

Complete order process including checkout, order confirmation, and status tracking.



Frontend Display

Product listing page (with filtering, search, pagination), product detail page, and related product recommendations.

02 System Architecture

The project follows Django's best practices, adopting a modular App structure to decouple different business functions.

Project Directory Structure

```
shop/ —┬─ accounts/ (User Accounts) — carts/ (Shopping
      | Carts)
      └─ products/ (Product Mgmt) — pages/ (Static Pages) —
orders/ (Orders)
      └─ static/ (Static Assets) — templates/ (HTML Templates)
      └─ manage.py (CLI Tool) & requirements.txt (Dependencies)
```

Core Data Flow

- 1. User Browsing:**User visits the pages or products module.
- 2. Product Selection:**User selects products and adds to carts.
- 3. Checkout:**User proceeds to checkout in the orders module.
- 4. Order Processing:**Orders module creates orders & calls payment.
- 5. User Center:**User manages info in users/accounts module.

Core Advantages:Modular design ensures independent function evolution, and a clear data flow loop guarantees business logic integrity and stability.

02 System Architecture (site map)

Level 1 Page	Level 1 Link	Level 1 App	Level 1 Free Text	Level 2 Page	Level 2 Link	Level 2 App	Level 2 Free Text	Level 3 Page	Level 3 Link	Level 3 App	Level 3 Free Text
Home	/	pages:index									
About	/about	pages:about	vision								
Product List	/products	products:index		Product Page	/products/<product_id>	products:product	product image name selections price description recommend products customer reviews	recommend product list	/products/<product_id>	products:product	product image name selections price description recommend products reviews
								customer review	#	n/a	
								 add to favorties	#	accounts:favor	
								 add to cart	#	cart:add_cart	
								cart	/cart	cart:index	
				 add to favorties	#	accounts:favor					
				 add to cart	#	cart:add_cart					
				pagination	/products?page=	products:product					
				filter	/products?{parameter}=	products:search					

02 System Architecture (site map)(cont.)

Level 1 Page	Level 1 Link	Level 1 App	Level 1 Free Text	Level 2 Page	Level 2 Link	Level 2 App	Level 2 Free Text	Level 3 Page	Level 3 Link	Level 3 App	Level 3 Free Text
Cart	/cart	cart:index	products image products name products amount products selection total price	 edit cart	#	cart:edit_cart					
				 delete cart	#	cart:delete_cart					
				 checkout	/checkout	checkout:index		checkout form	/checkout/form		payment method, shipping method email
				product page	/products/<product_id>	products:product	product image name selections price description recommend products customer reviews				
Search	/products?{parameter}=	products:search		Product List	/products	products:index					

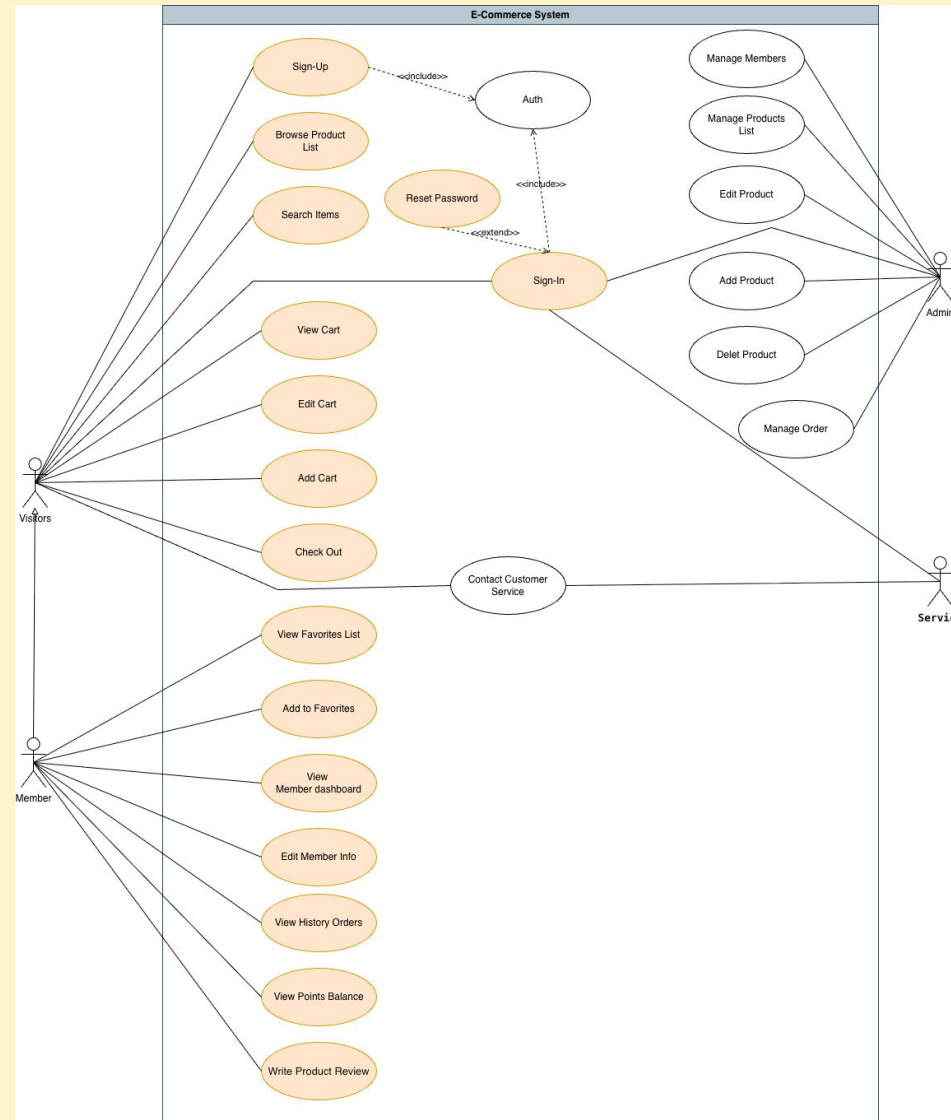
02 System Architecture (site map)(cont.)

Level 1 Page	Level 1 Link	Level 1 App	Level 1 Free Text	Level 2 Page	Level 2 Link	Level 2 App	Level 2 Free Text	Level 3 Page	Level 3 Link	Level 3 App	Level 3 Free Text
Account	/accounts	accounts:dashboard	display user name display member number display point balance	user info	/accounts/info	accounts:info	display email display name display phone (encrypt) password	edit information : email, phone, password, username	#	accounts:edit_info	
≡				favorties list	/accounts/favorties	accounts:favorties	display favorties	 delete favorties	#	accounts:delete_favorties	
								 add cart	#	cart:add_cart	
				order history list	/accounts/orders	accounts:orders		History order	/orders/< order.id >	orders:order	rate review message
Sign In	/signin	accounts:signin	email input field password input field OAuth	Sign up	/signup	accounts:signup					
				forgot / reset password	/reset_password	accounts:reset_password					

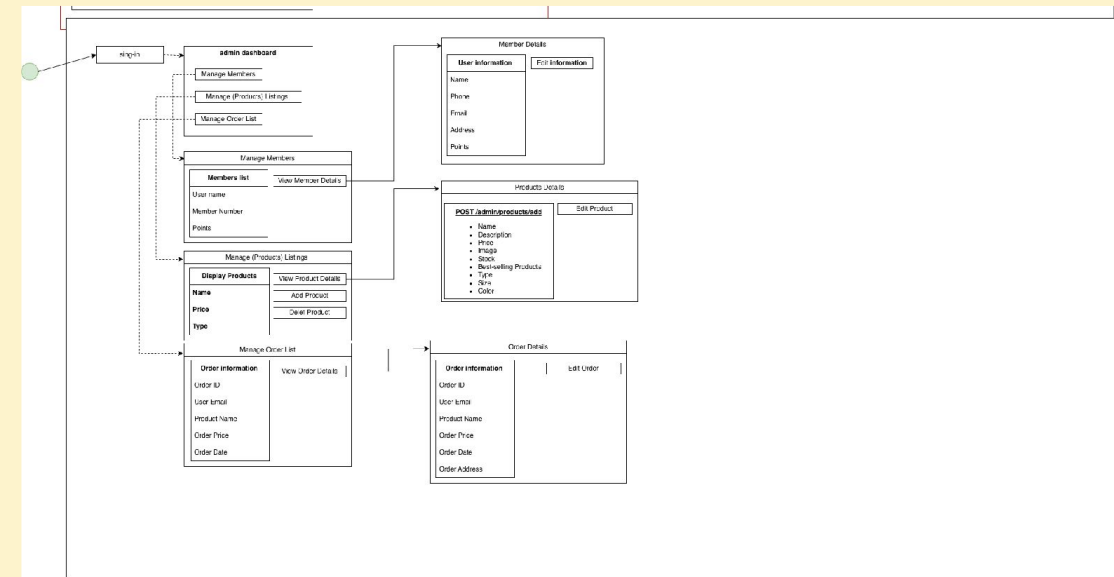
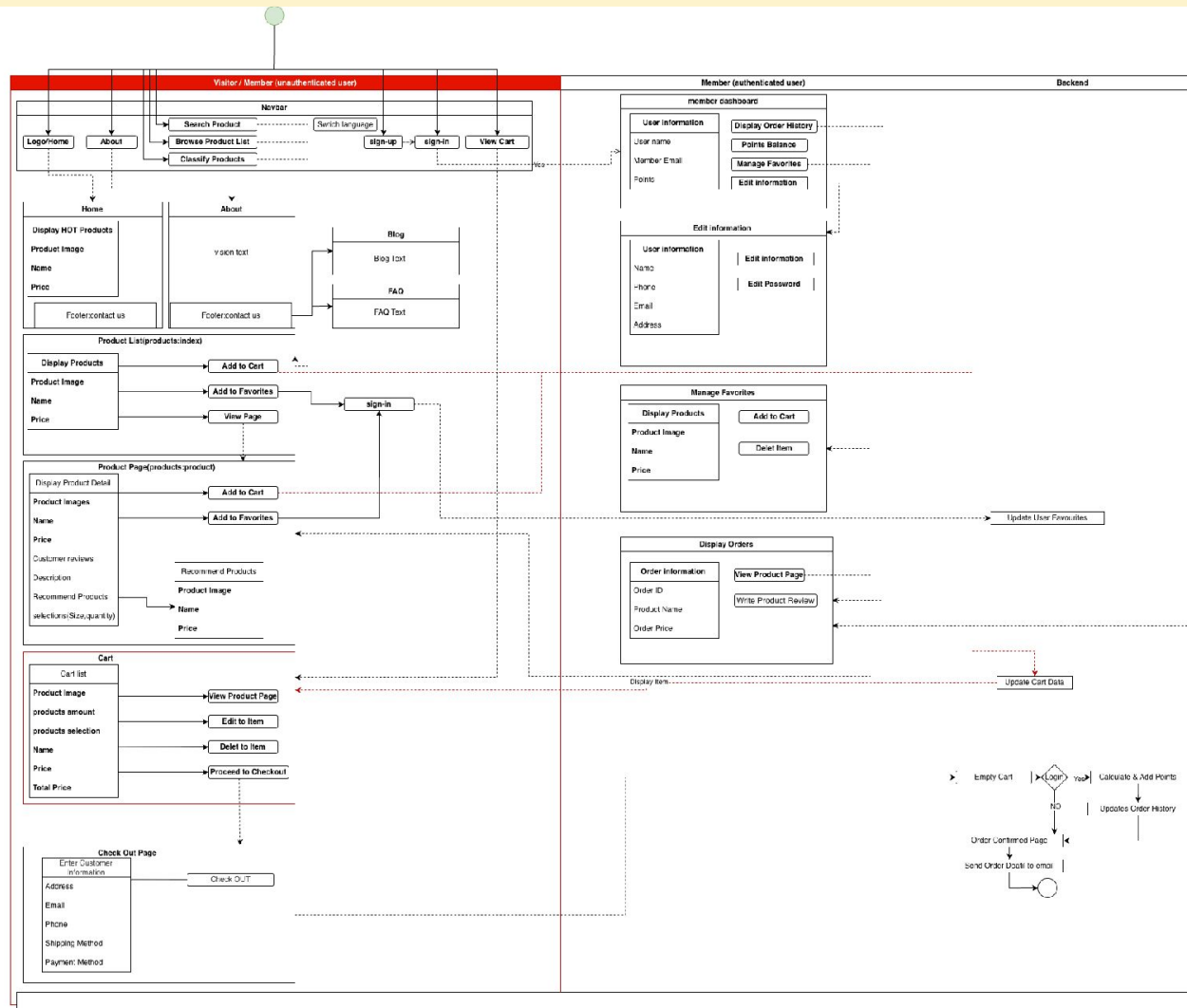
02 System Architecture (site map)(cont.)

Level 1 Page	Level 1 Link	Level 1 App	Level 1 Free Text	Level 2 Page	Level 2 Link	Level 2 App	Level 2 Free Text	Level 3 Page	Level 3 Link	Level 3 App	Level 3 Free Text
Blog	/blogs	blogs:index		 pagination	#						
FAQ	/faq	pages:faq		Contact us							
Payment Info	/payment-info	pages:payme nt_info									
Return Info	/return-info	pages:return _info									
Shipping Info	/shipping-info	pages:shippi ng_info									
Contact Us	/contact	contacts:cont act		email us	#	contacts:messa ge					
language	#	i18n	Chinese or English								
light mode	#	i18n	light or dark								
logo	/	pages:index	return Home Page								

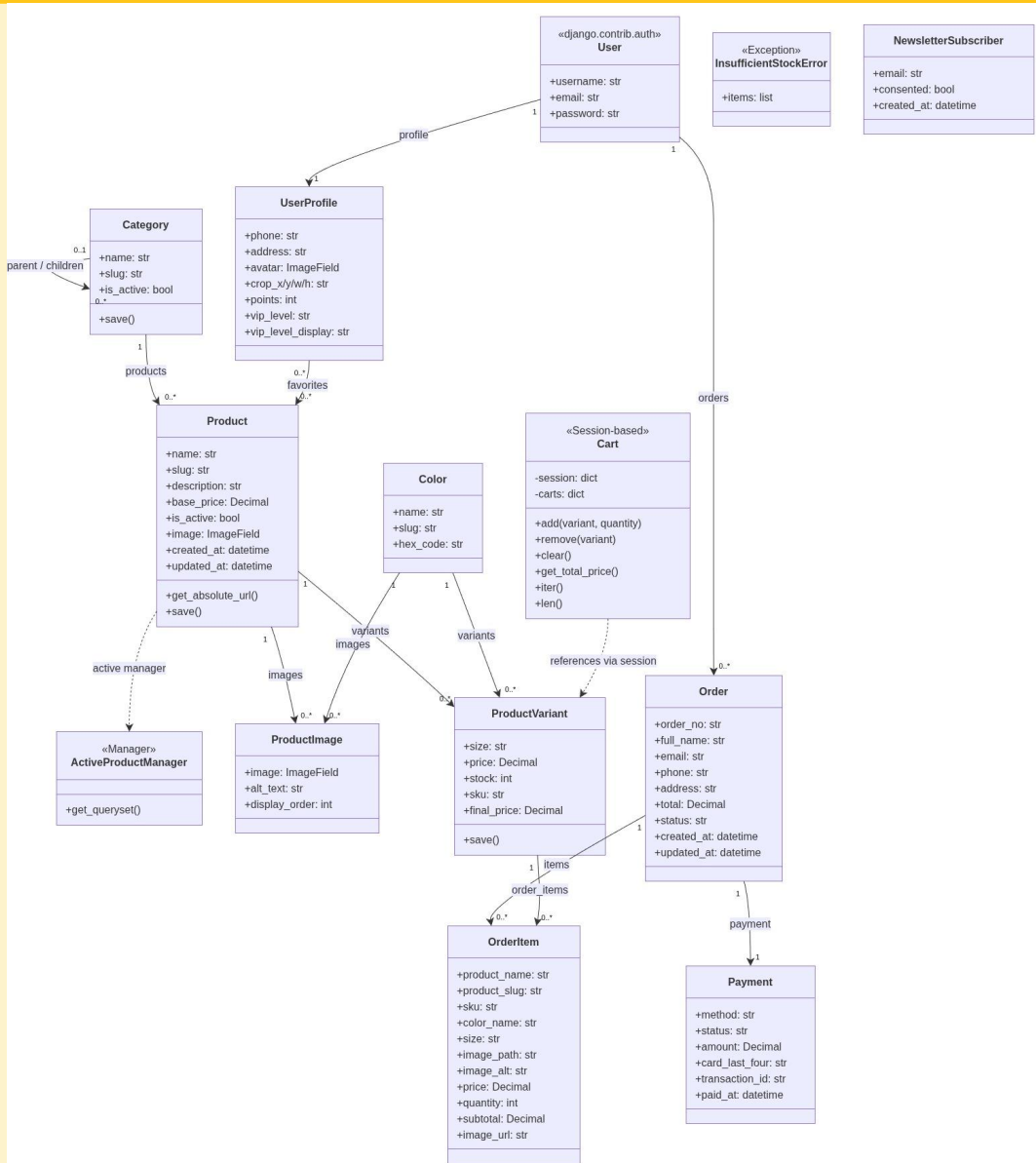
02 System Architecture (Use Case)



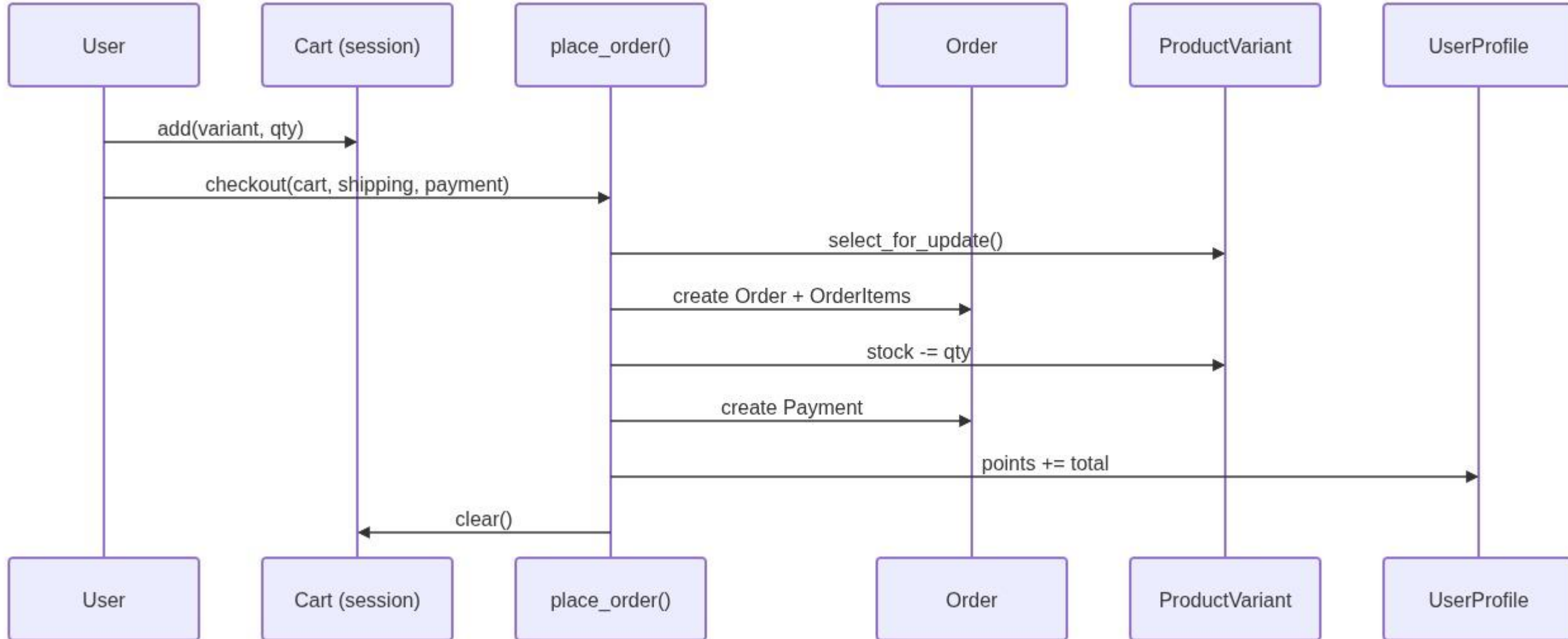
02 System Architecture (Sequence Diagram | Purchase)



03 System Architecture (Class Diagram)



04 System Architecture (Sequence Diagram | Checkout)



03 Core Module Details - Product Management (`products`)

This module is the core of the e-commerce system, and its design demonstrates a deep understanding of complex product structures.

Category

Implements a tree structure to support unlimited category levels through a self-referential `parent` foreign key.

Product

The main product table stores basic information. It features a custom manager `ActiveProductManager` that only queries active products by default.

Variant (Core)

Uniquely identifies a sellable unit by combining `Product`, `Color`, `Size`. Each has independent `stock` / `price` and auto-generates SKU on save.

Color & Image

Supports binding different images to products of different colors, enhancing the user experience.

Core Idea: The **`ProductVariant`** model elegantly solves multi-spec product management, establishing a flexible and scalable system.

Thank You



Q & A